

Software Requirements Specification (SRS)

for

Telemedicine & Remote Patient Monitoring (RPM) System

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Software Requirements Specification

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the requirements for the Telemedicine & Remote Patient Monitoring (RPM) System. This document serves as the foundation for design, development, testing, and deployment. It defines the functional capabilities, external interfaces, performance benchmarks, and security constraints necessary to deliver a reliable and HIPAA-compliant healthcare solution.

1.2 Document Conventions

- **Must:** Indicates a mandatory requirement essential for system operation.
- **Should:** Indicates a recommended requirement that adds value but is not critical.
- **May:** Indicates an optional requirement.
- **Shall:** Indicates a binding contractual requirement.
- **R#:** Unique requirement identifier for traceability.

1.3 Intended Audience

- **Development Team** — implement the system according to specifications.
- **Project Managers** — plan timelines, resources, and scope.
- **Quality Assurance (QA) Team** — create test plans, test cases, and validation protocols.
- **Clinical Stakeholders (Doctors/Nurses)** — validate clinical workflow needs.
- **Regulatory Compliance Officers** — ensure HIPAA and GDPR compliance.

1.4 Product Scope

The RPM System is a cloud-based web and mobile platform designed to bridge the gap between patients and healthcare providers. It will monitor patients remotely through approved IoT wearable devices; enable secure high-definition virtual consultations; provide centralized EHR and vital-sign dashboards; and automatically generate SMS, push, and email alerts when vitals exceed configured thresholds.

The system aims to reduce hospital readmission rates by 30% and improve chronic disease management for diabetes, hypertension, and COPD through proactive intervention.

1.5 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
- HIPAA Privacy Rule (45 CFR Part 160 and Subparts A and E of Part 164).
- GDPR Regulation (EU) 2016/679.
- FDA Guidance on Medical Device Data Systems (MDDS).
- System Business Requirements Document (BRD) v2.3.

2. Overall Description

2.1 User Needs

User Type	Needs Description
Patients (Chronic Disease)	Intuitive mobile app to connect wearables, view health metrics, schedule virtual appointments, and receive medication reminders.
Physicians/Doctors	Web clinical dashboard for real-time trends, historical data, critical alerts, and video consultations.
Nursing Staff/Triage	Monitor alert queues, triage emergency alerts, and escalate critical cases.
System Administrators	Manage accounts, device integrations, alert thresholds, and compliance audit logs.
Family Members/Caregivers	Limited view-only access to vital signs with patient consent.

2.2 Assumptions and Dependencies

2.2.1 Assumptions

- A-01: Patients have stable Wi-Fi or 4G/5G connectivity.
- A-02: Patients can use iOS/Android smartphones and pair Bluetooth wearables.
- A-03: IoT wearable devices provide standardized BLE or RESTful APIs.
- A-04: The healthcare organization has a secure VPN for internal EHR access.

2.2.2 Dependencies

- D-01: Twilio API for SMS alerts and SendGrid API for transactional email.
- D-02: AWS Cloud (EC2, RDS, S3, Lambda) for hosting and scalability.
- D-03: Epic EHR integration depends on hospital API credentials and data mapping specifications.

3. System Features and Requirements

3.1 Functional Requirements

3.1.1 User Authentication & Profile Management

- R-101: Register patients and doctors using email, phone number, and strong password (minimum 8 alphanumeric characters).
- R-102: Support RBAC with Patient, Physician, Nurse, and Admin roles.
- R-103: Enforce MFA via OTP for all doctor and admin accounts.
- R-104: Allow patients to update address, emergency contact, allergies, and medications.

3.1.2 IoT Wearable Device Integration & Data Ingestion

- R-201: Allow patients to pair up to 3 wearable devices via BLE or device ID.
- R-202: Ingest vital signs every 5 minutes (configurable) and store them in a time-series database.
- R-203: Support Heart Rate, SpO2, Blood Pressure, Blood Glucose, Body Temperature, and Respiratory Rate.
- R-204: If no device data is received for more than 60 minutes, alert the patient and assigned nurse.

3.1.3 Clinical Dashboard & Alerting

- R-301: Provide a web dashboard listing assigned patients sorted by Critical, Warning, Stable.
- R-302: Allow doctors to configure patient-specific warning and critical thresholds.
- R-303: Trigger Critical Alert with red banner, audible alarm, SMS, and push notification when thresholds are crossed.
- R-304: Trigger Warning Alert with yellow banner and push notification.
- R-305: Display 24-hour, 7-day, and 30-day vital-sign trends using line charts.

3.1.4 Virtual Consultations

- R-401: Allow doctors to schedule video consultations with unique meeting links.
- R-402: Allow patients to request an Urgent Consult that enters the triage queue.
- R-403: Encrypt video consultations end-to-end using WebRTC technology.
- R-404: Record consultations only with explicit consent from both parties and retain securely for 7 years.

3.1.5 Electronic Health Records (EHR) Management

- R-501: Allow doctors to document SOAP clinical notes.
- R-502: Allow PDF prescriptions to be sent electronically to the patient's preferred pharmacy.
- R-503: Maintain an immutable EHR audit trail containing timestamp, user ID, and before/after values.

3.1.6 Reporting & Analytics

- R-601: Generate monthly Patient Health Summary PDFs.
- R-602: Generate Alert Effectiveness reports measuring responses within the 5-minute SLA.
- R-603: Provide population health analytics such as diabetic patients with HbA1c above 7.0%.

3.2 External Interface Requirements

3.2.1 Hardware Interfaces

- HI-01: Interface with FDA-cleared Bluetooth wearables through manufacturer SDKs or standard GATT over BLE.
- HI-02: Support hospital bedside monitors such as Philips IntelliVue via HL7 where available.

3.2.2 Software Interfaces (APIs)

- SI-01: Integrate with Epic EHR via HL7 FHIR for patient demographics and clinical notes.
- SI-02: Interface with Twilio Programmable SMS API for critical alerts.
- SI-03: Interface with SendGrid Email API for reminders, health summaries, and password resets.

3.2.3 User Interfaces

- UI-01: Patient mobile app must show Today's Health with current vitals, medication reminders, and appointments.
- UI-02: Doctor dashboard must show color-coded Stable, Warning, and Critical patient status.
- UI-03: Nurse triage queue must sort alerts by severity/timestamp and provide Acknowledge and Escalate to Doctor actions.

3.3 Non-Functional Requirements

3.3.1 Performance & Scalability

- NFR-01: Ingest data from at least 10,000 concurrent devices.
- NFR-02: Load a 7-day vital-sign trend chart in under 2 seconds.
- NFR-03: Deliver critical alerts to doctor app and SMS within 5 seconds.
- NFR-04: Establish video sessions within 3 seconds at minimum 720p HD.

3.3.2 Security & Compliance

- NFR-05: Encrypt PHI at rest with AES-256 and in transit with TLS 1.3.
- NFR-06: Maintain detailed patient-record access audit logs.
- NFR-07: Enforce 10-minute inactivity timeout for clinical users.
- NFR-08: Support GDPR Article 17 deletion requests subject to legal retention requirements.

3.3.3 Reliability & Availability

- NFR-09: Guarantee 99.95% uptime for alerting and vital-sign ingestion 24/7.
- NFR-10: Back up PostgreSQL every 4 hours and fail over to a secondary AWS region within 30 minutes.
- NFR-11: Cache vital-sign data locally for up to 24 hours during network outages and auto-sync afterward.

3.3.4 Usability

- NFR-12: Design for low technical proficiency with large fonts, high contrast, and voice-over support.
- NFR-13: Support English, Spanish, and Mandarin.
- NFR-14: 90% of new patients should pair a wearable and view vitals within 5 minutes without support.

3.3.5 Maintainability

- NFR-15: Containerize backend services with Docker and orchestrate using Kubernetes.
- NFR-16: Maintain at least 85% unit-test coverage and integration tests for third-party APIs.
- NFR-17: Structure logs as JSON and aggregate them into ELK for monitoring and troubleshooting.

4. Appendixes

Appendix A: Data Dictionary

Data Element	Data Type	Description	Constraints
Patient_ID	UUID	Unique identifier for patient.	Primary Key, auto-generated.

Data Element	Data Type	Description	Constraints
Device_ID	VARCHAR(100)	MAC address or serial number of wearable.	Unique per patient.
Vital_Sign_Type	ENUM	HR, SpO2, BP, Glucose, Temp, Resp.	Predefined, mandatory.
Vital_Value	DECIMAL(10,2)	Numerical device reading.	Physiologically possible range.
Alert_Threshold_Critical	DECIMAL(10,2)	Critical threshold set by doctor.	Must exceed warning threshold.
Consultation_Record_URL	VARCHAR(500)	S3 location for recorded video.	Nullable, encrypted.
Is_Consent_Given	BOOLEAN	Whether patient consented to sharing/recording.	Defaults FALSE.

Appendix B: Acronyms and Abbreviations

Acronym	Full Form
RPM	Remote Patient Monitoring
PHI	Protected Health Information
EHR	Electronic Health Record
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection Regulation
HL7	Health Level Seven (Interoperability Standard)
FHIR	Fast Healthcare Interoperability Resources
IoT	Internet of Things
BLE	Bluetooth Low Energy
SOAP	Subjective, Objective, Assessment, Plan
SLA	Service Level Agreement
MFA	Multi-Factor Authentication
AWS	Amazon Web Services

End of Document